

Problem Set 4: Elasticity

Questions and Solutions

- 1) In 2028, the price was \$9.50 and 9 units were supplied. In 2029, the price increased to \$10.50 and 11 units were supplied. What is the own-price elasticity of supply?

Solution: Elasticity of supply is

$$\varepsilon_S = \frac{\% \Delta Q^S}{\% \Delta P}$$

When you have two points with which to calculate elasticity, use the mid-point method. This would make

$$\% \Delta P = \frac{10.5 - 9.5}{\frac{10.5 + 9.5}{2}} = \frac{1}{10}$$

and

$$\% \Delta Q^S = \frac{11 - 9}{\frac{11 + 9}{2}} = \frac{2}{10}$$

Putting them together gives us

$$\varepsilon_S = \frac{\% \Delta Q^S}{\% \Delta P} = \frac{\frac{2}{10}}{\frac{1}{10}} = 2$$

Supply is elastic: a 1% change in price leads to a 2% change in quantity supplied.

- 2) For a 2% decrease in quantity, what is the % change in price that causes it, for each of the (own) price elasticities below?

2a) Own-price elasticity of demand is 2.

Solution: Since $|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right|$, plugging in the values gives us

$$|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right| \Rightarrow 2 = \left| \frac{0.02}{\% \Delta P} \right| \Rightarrow |\% \Delta P| = 0.01 = 1\%$$

Note: Why am I writing 0.02 instead of just 2? Because it is the technically correct way to write a percentage. As a shortcut we sometimes write 2, and in a question like this, where it is a ratio of percentages, it doesn't matter. It does matter elsewhere, though. For example: if the price goes up by 2%, what is the increase in price? If you calculate the increase in price as $P * 2$ you will be wrong, because you need to use $P * 0.02$. **So a 1% increase in price produces the 2% decrease in quantity demanded.**

2b) Own-price elasticity of demand is 1.

Solution: Following the same logic as in part (a),

$$|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right| \Rightarrow 1 = \left| \frac{0.02}{\% \Delta P} \right| \Rightarrow |\% \Delta P| = 0.02 = 2\%$$

So a 2% increase in price produces the 2% decrease in quantity demanded.

2c) Own-price elasticity of demand is 0.5.

Solution: Following the same logic as in part (a),

$$|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right| \Rightarrow 0.5 = \left| \frac{0.02}{\% \Delta P} \right| \Rightarrow |\% \Delta P| = 0.04 = 4\%$$

So a 4% increase in price produces the 2% decrease in quantity demanded.

- 3) Assume a 5% increase in price.

3a) If elasticity of demand is 5, what is the resulting change in quantity demanded?

Solution: This is a similar type of question to the one above, except that since you aren't given starting or ending prices, you can only solve for the resulting percentage change in Q^D . Although we only solve for absolute values of own-price demand elasticity, we know that in 101 we are only looking at downward sloping demand. That means that a complete answer will include not only the percentage change in Q^D but the direction of change as well.

$$|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right| \Rightarrow 5 = \left| \frac{\% \Delta Q^D}{0.05} \right| \Rightarrow |\% \Delta Q^D| = 0.25$$

When prices increase by 5%, Q^D decreases by 25%

3b) If elasticity of demand is 1.5, what is the resulting change in quantity demanded?

Solution: $|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right| \Rightarrow 1.5 = \left| \frac{\% \Delta Q^D}{0.05} \right| \Rightarrow |\% \Delta Q^D| = 0.075$

When prices increase by 5%, Q^D decreases by 7.5%

4) For each statement below tell us if it is True/False/Uncertain (TFU) and briefly explain your reasoning.

4a) You own a legal marijuana store in Ontario. The price of 1 gram of your product falls from \$15 to \$12. Your revenue falls when this happens. TFU: demand for your marijuana is inelastic.

Solution: True. A change just in price is a movement along the demand curve. A decrease in price causes an increase in quantity demanded. If the quantity demanded rises by a smaller % than the % fall in price, revenues fall. Here, prices fell and revenues decreased. **It must be because the % increase in quantity demanded is less than the % decrease in price, which is the definition of inelastic demand.**

4b) Let's assume that the food trucks outside Robarts operate in a perfectly competitive market. They all sell identical shawarmas and are price takers. You own one of these trucks. TFU: The demand for your products is perfectly inelastic.

Solution: False. Since all the trucks sell shawarmas that are identical to consumers, if you increase your prices all your customers will just move to another truck. Since it is assumed to be a perfectly competitive market, your customers are able to find all the shawarmas they need. If you decrease your price relative to the others, you get (and by assumption are able to satisfy) all the consumers looking to buy a shawarma. **This makes the demand you face perfectly elastic.** A slight change in price leads to a dramatic change in the quantity of your goods demanded.

4c) You own a gas station. The gas is delivered once a month and you can't get any more until the next month. TFU: Your supply of gas on the 15th of the month is perfectly inelastic.

Solution: True. No matter how much the price rises, you can't supply more gas. Even if the price falls, you're not going to supply any less. Basically, the change in quantity is zero for any % change in price. **As the numerator in the elasticity formula is zero, your supply elasticity is zero, and hence perfectly inelastic.** (Note: this implicitly assumes you always put all your inventory for sale regardless of price).

5) The demand curve is captured by $Q^D = 450 - 3P$.

5a) Calculate own-price elasticity of demand at $P = 30$ and $P = 90$, in each case indicating whether demand is elastic or inelastic.

Solution: Elasticity is $|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right|$ So far we have been calculating the % change using the mid-point method. For cases when we are given a demand equation we can do it another way.
Step 1: Let's take a closer look at what a % change is

$$\% \Delta Q^D = \frac{\text{change}}{\text{original}} = \frac{\Delta Q^D}{Q^D}$$

We used the mid-point method when given two points because we don't want the starting point to matter. With two points we calculated the point in the middle (i.e. the average) to use in the denominator to address this. When we have an equation for demand, our lives are easier, as we can bring the two extremes used to calculate the average closer and closer together until they are exactly one point. This way the starting point

doesn't matter, as we are calculating elasticity at a particular point.

Step 2: Rewrite the elasticity formula as

$$|\varepsilon_D| = \left| \frac{\% \Delta Q^D}{\% \Delta P} \right| = \left| \frac{\frac{\Delta Q^D}{Q^D}}{\frac{\Delta P}{P}} \right| = \left| \frac{\Delta Q^D}{\Delta P} \frac{P}{Q^D} \right|$$

Step 3: From the skills checked on the Math assessment, I know that you know how to figure out the change in Q^D (i.e. ΔQ^D) for a given change in P (i.e. ΔP) given an equation. For this demand curve $Q^D = 450 - 3P$ you can see that

$$\frac{\Delta Q^D}{\Delta P} = -3$$

Plugging this back into the elasticity equation, we get

$$|\varepsilon_D| = \left| \frac{\Delta Q^D}{\Delta P} \frac{P}{Q^D} \right| = \left| -3 * \frac{P}{Q^D} \right|$$

Step 4: Find the elasticity at the point you are looking for.

Elasticity at $P = 30 \Rightarrow Q^D = 360$ is

$$|\varepsilon_D| = \left| -3 * \frac{P}{Q^D} \right| = \left| -3 * \frac{30}{360} \right| = \frac{1}{4}$$

which makes demand inelastic.

Elasticity at $P = 90 \Rightarrow Q^D = 180$ is

$$|\varepsilon_D| = \left| -3 * \frac{P}{Q^D} \right| = \left| -3 * \frac{90}{180} \right| = \frac{3}{2}$$

which makes demand elastic.

Notice that this is a linear demand curve, but elasticity varies along the curve. Make sure you can understand and explain why this is the case.

5b) What price maximizes total revenue?

Solution: Total revenue is maximized along a linear demand curve at the point where we have unit elasticity, i.e. $|\varepsilon_D| = 1$. You can find this point in two ways.

Method 1: Use the fact that the unit elastic point is exactly halfway along the curve. The intercept on the P axis is $Q^D = 450 - 3P = 0 \Rightarrow P = 150$. **This makes the halfway price point $P = \frac{150}{2} = 75$.**

Method 2: Use the elasticity formula from the previous parts to find the point where $|\varepsilon_D| = 1$

$$\begin{aligned} |\varepsilon_D| &= \left| -3 * \frac{P}{Q^D} \right| \\ 1 &= 3 * \frac{P}{450 - 3P} \\ 450 - 3P &= 3P \\ P &= \frac{450}{6} \\ P &= 75 \end{aligned}$$

5c) Is the price you found in part (b) the price that maximizes your total profit?

Solution: No. The price we found $P = 75$ maximizes total revenue. Profit is Revenue minus Total Costs. Depending on what the costs look like there may be another price that maximizes total profit.

However, if marginal cost were always zero, then that price would also maximize total profits. Why? Every extra unit costs you zero to make, so any extra revenue from that unit just raises profits.

6) Below is the abstract from the paper “[To Borrow or Not to Borrow: Religious Norms and the Elasticity of Demand for Credit](#)” by Dean Karlan, Adam Osman and Nour Shammout. Feb 2017.

“Low utilization of credit in developing countries may be partially due to societal norms. We consider one such case in Jordan and compare the demand for a new, sharia-compliant product to a non-compliant product. To comply with the Islamic prohibition on paying or receiving interest, the sharia-compliant product uses a bank fee rather than interest payment structure, while keeping the economics of the product the same as a comparable conventional loan. We find that in this largely Muslim country, consumers offered a sharia-compliant loan increase their application rate from 18% to 22%. We also randomly varied the price of the sharia-compliant product, and find that the less religious individuals in our sample are twice as elastic with respect to price as those who are more religious. We find no evidence of differences between those who apply for the conventional loan and those that apply for the sharia-compliant loan on observable demographics, suggesting that this new product successfully increases utilization of formal financial services without necessarily pulling in more risky individuals”

- 6a) What could be one reason why religious individuals have a lower (absolute) price elasticity of demand for sharia compliant loans?

Solution: When the price rises, quantity demanded falls as people shift their consumption to different alternatives/substitutes (or do without, which may be more difficult depending on the situation and the good). In this case, if sharia-compliant loans are important for religious individuals, then they have fewer other options to switch to when the price of a loan rises. **This lack of alternatives means that their demand for sharia-compliant loans will fall more slowly than that of similar non-religious individuals, who have more options.**

- 6b) In general, across all individuals, would you expect the elasticity of demand for sharia-compliant loans to be more or less elastic than the demand for any loans?

Solution: (Own price) demand elasticity (in absolute terms) is, in general, bigger for narrower categories. **As sharia-compliant loans are a subset of all loans, demand is likely to be more elastic for them than for all loans.** If the price of sharia-compliant loans increases, people can switch to other loans (albeit reluctantly in the case of religious individuals). However, if the price of all loans (which includes sharia-compliant loans) increases, there are fewer alternatives if an individual needs funds, making it less likely they will substitute away.

- 7) Preventive health products with public health benefits (e.g. insecticide-treated bednets, water purification products, and latrine slabs for safe sanitation) have a high price sensitivity in developing countries (i.e. high absolute own-price demand elasticity). See “[The Price is Wrong](#)” J-PAL Bulletin. April 2017. Cambridge, MA: Abdul Latif Jameel Poverty Action Lab.

- 7a) If you are selling insecticide-treated bednets to prevent malaria, would raising the prices increase or decrease revenues for your company?

Solution: We are always looking at downward sloping demand in Eco 101. With elastic demand, when the price increases, the more-than-proportional decrease in quantity causes revenues to decrease, so **raising prices would lower revenues for the company.**

- 7b) Given the importance of health and the big impact these health interventions have, what could be the reasons why people are so price sensitive even at very low prices?

Solution: A high price sensitivity means that for a given increase in price, people reduce the quantity they buy more than proportionally. In the earlier question, we talked about elasticity depending on the alternatives and how easy it was to switch to them. For this example, it may be a case of people choosing to do without when the price rises.

Preventive health measures reduce disease incidence in the future. **That means not buying it now has no discernible effect on health immediately, only in the long run, so it is possible to do without.** This is different when, for example, you’re sick and need antibiotics or surgery. That is much harder to do without.

Paying for the health intervention today means people need to consider the implicit cost of the cash. If they didn’t use the cash for the health measure, what else could they have spent it on? In a developing country, having enough cash on hand to pay for the intervention may come at the cost of having cash to pay for necessities like food.

Some studies find that when people were given cash or loans to buy the health intervention, or given time to

raise the funds, they were willing to pay more. You can read more details on this on page 8 of “[The Price is Wrong](#)” J-PAL Bulletin. April 2017. Cambridge, MA: Abdul Latif Jameel Poverty Action Lab.

Checked with Claude (Anthropic) for grammar, clarity and errors. Tell me if you find one I missed.